

Hitec University Taxila, Department of Computer Engineering

Lab Manual for Mobile Application Development

Lab-13 Android Broadcast Receiver

Contents

1	Introduction	3
2	Activity Time boxing	3
3	Objectives of the experiment.....	3
4	Walk through Task	3

1 Introduction

Broadcast Receivers simply respond to broadcast messages from other applications or from the system itself. These messages are sometime called events or intents

2 Activity Time boxing

Table 1: Activity Time Boxing

Task#	Activity Name	Activity time	Total Time
1	Introduction	60mins	60min
2	Walk Through Task	60mins	60min
3	Evaluation Task	60min	60min

3 Objectives of the experiment

- To get basic understanding regarding android broadcast receiver
- To get basic understanding regarding how to use android broadcast receiver
- To get to know about the benefits related to android broadcast receiver

4 Walk through Task

Creating New Activity

Step 1: Create Activities:

Create a new project just like you have created in your previous labs and create one more empty activity.

Step 2: Design XMLs of Activity:

Designs of activity are given below, just use the given xml code which is given in this step.

activity_main_broadcast_recv.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="sendBR"
        android:text="Send BR" />
</LinearLayout>
```

Step 3: Write MyReceiver class

```
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.widget.Toast;

public class MyReceiver extends BroadcastReceiver {
    @Override
```

```

        public void onReceive(Context context, Intent intent) {
            Toast.makeText(context, "Intent Detected: " + intent.getAction(),
                Toast.LENGTH_LONG).show();
        }
    }
}

```

Step 4: Write “MainBroadcastRecv.java” class

```

import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;
import android.view.View;

import androidx.appcompat.app.AppCompatActivity;

public class MainBroadcastRecv extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main_broadcast_recv);

        MyReceiver receiver = new MyReceiver();
        IntentFilter filter1 = new
        IntentFilter("android.intent.action.AIRPLANE_MODE");
        registerReceiver(receiver, filter1);

        IntentFilter filter2 = new
        IntentFilter("com.example.mybroadcastreceiver.CUSTOM_INTENT");
        registerReceiver(receiver, filter2);

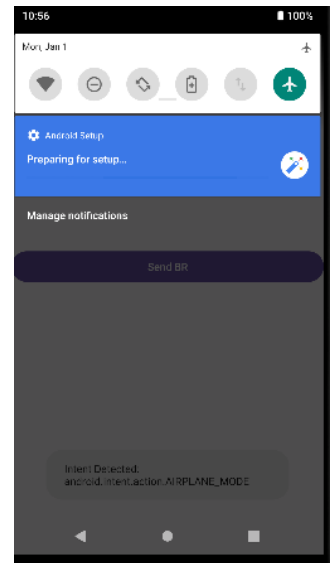
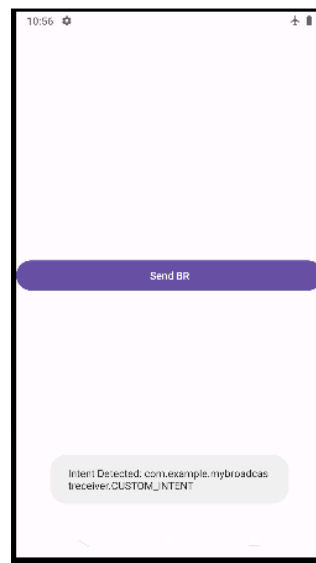
    }

    public void sendBR(View view) {
        Intent intent = new
        Intent("com.example.mybroadcastreceiver.CUSTOM_INTENT");
        sendBroadcast(intent);
    }
}

```

Step 6: Now Run Project

Click “Run” or press “Shift + F10”.



Step 6: Home Task

Consider previous browser app that you have designed in Lab task # 12, now use two broadcast events i.e. connectivity change, and WiFi connectivity, update the app to detect those events and display message to the user if connectivity and or WiFi status is changed using toast message, extra marks will be given for whom those use background or status icon to show if connectivity and/or WiFi is enable or not.